Tenkasi District

Common Quarterly Examination - 2023

25-09-2023.

Standard 12

CHEMISTRY

Part - I

Time: 3.00 Hours Marks: 70

15x1=15

**Answer all the questions:**

1) Which one of the following reaction represents calcinations?

a) $2Zn + O_2 \rightarrow 2ZnO$ b) $2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$

c) $MgCO_3 \rightarrow MgO + CO_2$ d) Both (a) and (c)

2) **Assertion :** Aqueous solution of potash Alum is acidic

**Reason :** Aluminium sulphate undergo hydrolysis.

a) Both assertion and reason are true and reason is the correct explanation of assertion

b) Both assertion and reason are true but reason is not the correct explanation of assertion

c) Assertion is true but reason is false

d) Both assertion and reason are false

3) Solid (A) reacts with strong aqueous NaOH liberating a foul smelling gas (B) which spontaneously burn in air giving smoky rings. A and B are respectively

a) $P_4$ (red) and $PH_3$ b) $P_4$ (white) and $PH_3$

c) $S_8$ and $H_2S$ d) $P_4$ (white) and $H_2S$

4) Phosgene is a common name for

a) phosphonyl chloride b) thionyl chloride

c) carbon dioxide and phosphine d) carbonyl chloride

5) A white crystalline salt (A) react with dilute HCl to liberate a suffocating gas (B) and also forms a yellow precipitate. The gas (B) turns potassium dichromate acidified with dil $H_2SO_4$ to a green coloured solution (C). A, B and C are respectively.

a) $Na_2SO_3$, $SO_2$, $Cr2(SO_4)_3$ b) $Na_2S_2O_3$, $SO_2$, $Cr_2$, $(SO_4)_3$

c) $Na_2S$, $SO_2$, $Cr_2(SO_4)_3$ d) $Na_2SO_4$, $SO_2$, $Cr_2(SO_4)_3$

6) The flame colours of metal ions are due to

a) Frenkel defect b) Schottky defect

c) metal deficiency defect d) metal excess defect

7) The radius of an atom is 300pm, if it crystallizes in a face centered cubic lattice, the length of the edge of the unit cell is

a) 488.5pm b) 848.5pm c) 884.5pm d) 484.5pm

8) For a reaction Rate = K [acetone]3/2 acetone then unit of rate constant and rate of reaction respectively is

a) $\left(molL^{-1}s^{-1}\right), \left(mol^{-1/2}L^{1/2}s^{-1}\right)$ b) $\left(mol^{-1/2}L^{1/2}s^{-1}\right), \left(molL^{-1}s^{-1}\right)$

b) $\left(mol^{1/2}L^{1/2}S^{-1}\right), \left(molL^{-1}S^{-1}\right)$ d) $\left(molLs^{-1}\right), \left(mol^{1/2}L^{1/2}s^{-1}\right)$

9) Half life for radioactive $^{14}C$ is 5760 yr. In how many years, 200 mg of $^{14}C$ will be reduced to 25 mg?

a) 5760 Yr b) 11520 Yr c) 17280 Yr d) 23040 Yr

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10) Conjugate base for Bronsted acids $H_2O$ and HF are

a) $OH^-$ and $H_2FH^+$ respectively b) $H_3O^+$ and $F^-$, respectively

c) $OH^-$ and $F^-$, respectively d) $H_3O^+$ and $H_2F^+$, respectively

11) $(CH_3)_3 - C - CH(OH)CH_3 \xrightarrow{Conc\,H_2SO_4} X$(major product)

a) $(CH_3)_3 CCH = CH_2$ b) $(CH_3)_2 C = C(CH_3)_2$

c) $CH_2 = C(CH_3)CH_2 - CH_2 - CH_3$ d) $CH_2 = C(CH_3) - CH_2 - CH_2 - CH_3$

12) Buffer solutions have constant acidity and alkalinity because

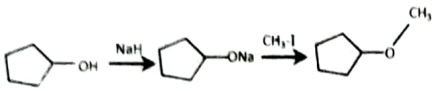
a) They have large excess of $H^+$ or $OH^-$ ions

b) They have fixed value of pH

c) These give unionized acid or base on reaction with added acid or alkali.

4) Acids and alkalies in these solutions are shielded from attack by other ions

13) The reaction




Can be classified as

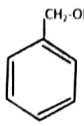
a) dehydration b) Williamson alcholsynthesis

c) Williamson ether synthesis d) dehydrogenation of alcohol

14) Predict the product Z in the following series of reactions

Ethanoic acid $\xrightarrow{PCl_5} X \xrightarrow[Anhydrous AlCl_3]{C_6H_6} Y \xrightarrow[ii)H_3O^+]{i)CH_3MgBr} Z$

a) $(CH_3)_2C(OH)C_6H_5$ b) $CH_3CH(OH)C_6H_5$

c) $CH_3CH(OH)CH_2 - CH_3$ d) 


15) Acetaldehyde is the rearrangement product of

a) Methyl alcohol b) Allyl alcohol

c) Vinyl alcohol d) Ethyl alcohol

## Part - II 6x2=12

**Answer any six of the following. (24 is compulsory):**

16) Explain Van arkel method

17) Explain the action of heat on Boric acid.

18) Write any three uses of silicones.

19) Classify the following solids

a) P4 b) Brass c) diamond d) NaCl

20) Give any three examples for the first order reaction.

21) Define solubility product.

22) What happens when glycerol react with $KHSO_4$?

23) Write the preparation of urotropine and its uses.

24) $K_b$ for $NH_4OH$ is $1.8 \times 10^{-5}$. Calculate the percentage of ionisation of 0.06M. ammonium hydroxide solution.

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## Part - III

**Answer any six of the following. (30 is compulsory):** 6x3=18

25) How will you prepare potash alum?

26) Calculate the number of unpaired electrons in $Ti^{3+}$, $Mn^{2+}$ and calculate the spin only magnetic moment.

27) What are the effects of lanthanide contraction?

28) Derive the integrated rate law for Zero Order reaction. A → product

29) Derive an expression for Ostwald's dilution law

30) Identify the order of the following reaction

a) Rusting of iron b) Radioactive decay of $_{92}U^{238}$.

c) $2A + 3B \rightarrow$ product, Rate = $k(A)^{1/2}(B)^2$.

31) Write three uses of diethyl ether

32) Write note on Benzoin Condensation

33) Identify the esters in the following fruits.

a) banana b) pineapple c) orange

## Part - IV

**Answer all of the following:** 5x5=25

34) a) i) What is Roasting? (2)

ii) Explain Cyanide leaching (3)

**(OR)**

b) i) Differeniate Diamond and graphite.

ii) What are interhalogen compounds? Explain types with examples.

35) a) Describe the preparation of $K_2Cr_2O_7$.

**(OR)**

b) i) Most of the transition metals and their compounds have catalytic activity. Why?

ii) What is Zeigler - Natta catalyst? Mention its uses.

36) a) i) Define Unit Cell (2)

ii) Calculate the packing efficiency of packing in case of face centerd cubic crystal (3).

**(OR)**

b) i) Explain briefly the collision theory of bimolecular reactions. (3)

ii) What are elementary reactions (2)

37) a) i) Derive Henderson equation (3)

ii) Calculate the pH of 0.04 M $HNO_3$ solution. (2)

**(OR)**

b) i) Convert glycol into a) ethylene b) epoxide c) dioxan

38) a) i) From Phenol how will you prepare

a) picric acid b) salicyilcacid c) azodye

**(OR)**

b) Explain the mechanism of aldol condensation.

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SIVAKUMAR.M,
Sri Ram Matric HSS
Vallam-627809
Tenkasi Dist.

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